Group 2 Project Brief: Community Blog Platform with AI Writing Assistant

Overview

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You will build a community blog platform, similar to a simple Medium: anyone can sign up, write and publish posts, and anyone can read them. An admin keeps the platform clean by handling reports. Inside the post editor, an AI writing assistant helps authors with titles, summaries, tags and clearer sentences, while the author stays in control of their own words. The goal is practice working as a team, not a perfect product. Aim for a working MVP in 3 weeks, using the same stack you used for your e-commerce project.

Who uses it

Role	What they can do
Visitor (not logged in)	Read posts, search, browse categories and author profiles
Author (any registered user)	Everything a visitor can do, plus write and publish posts, comment, like and report content
Admin	Manage categories, review reports, remove posts or comments and block users

MVP scope (must-have)

- Sign up and log in; every registered user is an author, and one or more accounts are admins
- Author writes a post with a title, body text, cover image, category and tags; saves it as a draft or publishes it directly
- Author edits or deletes their own posts and sees them on a "My posts" page
- Public homepage with the latest posts, category pages, a post page and an author profile page
- Search by keyword and filter by category or tag
- Logged-in users comment on and like posts
- Logged-in users report a post or comment; the admin sees reports and removes content or blocks the user
- Like and comments on blog posts
- AI writing assistant in the post editor (details below)

AI writing assistant: how it works

The editor has an "AI assistant" panel with four buttons. Each one sends the author's draft (or the selected paragraph) to an LLM API (such as Claude, OpenAI or Gemini) from the backend and shows the result as a suggestion:

Button	What the AI returns
Suggest titles	3 title options based on the post
Write a summary	A 2–3 sentence excerpt shown under the title on the homepage
Suggest tags	Up to 5 tags, matched to existing tags where possible
Improve this paragraph	A clearer version of the selected paragraph with grammar fixed, shown before and after

Ask the API for JSON only, for example:

```
{
  "titles": ["How Kigali's New Bus Lanes Changed My Commute", "One Month on Kigali's Bus Lanes", "Are the New Bus Lanes Working?"],
  "summary": "A daily commuter shares what changed after Kigali opened dedicated bus lanes on two main routes.",
  "tags": ["Kigali", "transport", "daily life"]
}
```

Rules the feature must follow:

- The author chooses. Nothing the AI suggests is saved until the author clicks Accept, and they can edit it first.
- The AI never writes a whole post. "Improve this paragraph" only fixes clarity and grammar and keeps the author's meaning and voice.
- The API key stays on the server and is never sent to the frontend.
- Each user gets a daily limit of AI requests (for example, 20), so one person cannot run up the API bill.
- If the AI call fails or takes too long, the editor shows a short message and the author keeps writing normally.
- Test the prompts on at least 10 sample drafts of different lengths and topics, and note where the suggestions were wrong or changed the author's meaning.

Stretch goals (only after the MVP works)

- AI moderation that checks new posts and comments for spam or abuse and sends flagged ones to the admin's report list
- AI translation of a post between English and Kinyarwanda, marked as machine-translated
- Following authors and a "Following" feed
- View count on each post

Out of scope: AI writing whole posts, paid subscriptions, ads, and private messaging.

Suggested feature ownership

Each person owns a feature end to end (database, API and screens). Swap roles if you prefer, but every member must own one.

Feature area	What it covers
Auth, profiles and admin	Sign up, log in, author profile pages, reports, removing content and blocking users
Writing posts	The post editor, image upload, drafts, publishing, editing and "My posts"
Public site and engagement	Homepage, category and post pages, search, filters, comments and likes
AI writing assistant	The AI panel, the LLM API calls and prompts, accepting suggestions, the daily limit, the test drafts

Shared by everyone: the database design, deployment and the README.

Timeline (3 weeks)

1. Week 1, plan and set up: agree on the screens (simple wireframes), the database tables and the API routes. Agree on the exact JSON each AI button returns, because the editor screen depends on it. Create the repo, the project board and all issues. Finish login and saving a draft by the end of the week; the AI owner gets a first working prompt in a small test script.
2. Week 2, build: each person builds their feature on their own branch, merging small pull requests often. Have one post written with AI suggestions, published, and showing on the homepage by Friday.
3. Week 3, finish: fix bugs, run the AI test drafts, test the full flow as a visitor, an author and an admin, deploy, write the README and rehearse the demo.

If you get a 4th week, use it for stretch goals and polish, not new core features.

Collaboration rules

- One shared GitHub repo with `main` protected. Nobody pushes to `main` directly.
- Work on feature branches (`feature/short-name`) and keep pull requests small.
- Every pull request needs at least one teammate's approval before merging. Reviewers leave real comments, not just an approval.
- Track all work as GitHub Issues on a project board, with one owner per issue.
- Hold a 15-minute standup every day: what I did, what I'll do next, what's blocking me.
- Never commit API keys; keep them in environment variables and list the variable names in the README.
- Show your progress to the trainers at the end of each week.

Deliverables

- Deployed link
- README with setup steps, environment variables, screenshots and who built what
- Demo accounts for two authors and an admin, with sample posts, comments and one report
- The AI test drafts and notes on how the suggestions performed

- A 5–10 minute demo in which every member presents their part, including writing a post with the AI assistant